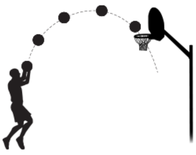
Parabolas

A quadratic equation has a variable that is squared, or raised to a power of 2. No variable in a quadratic equation can have an exponent greater than 2.

The graph of a quadratic equation forms a **parabola**, a symmetrical curved line. A parabola can open upward or downward.



When you throw a ball into the air, the path of the ball is a parabola.

To graph a quadratic equation, rewrite the equation so that one side equals 0. Then replace 0 with y.

Solve $x^2 = 4$.	You already know the equation has two answers: 2 and −2.
Subtract 4 from both sides.	$x^2 - 4 = 0$
Replace 0 with y.	$x^2 - 4 = y$, which can be written $y = x^2 - 4$

Now you are ready to graph the equation.

Example

Hide

Graph: $y = x^2 - 4$

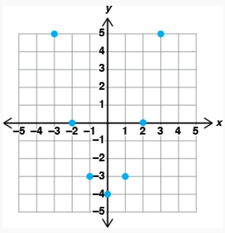
Step 1

Make a table of values. Choose values for x. Substitute them into the equation and solve for y.

x	y
−3	5
−2	0
−1	−3
0	−4
1	−3
2	0
3	5

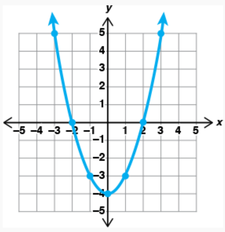
Step 2

Plot the points from the table.



Step 3

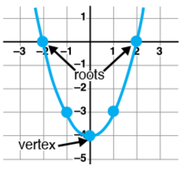
Sketch the parabola.



The graph tells important information about the equation.

The lowest or highest point on a parabola is its **vertex**.

Here, the vertex shows the lower limit of the range. The range of the equation is all real numbers greater than or equal to −4.

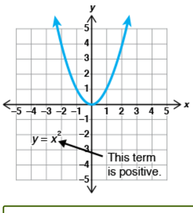


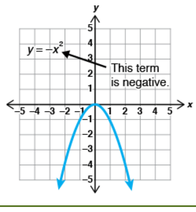
The **roots** are found where the graph crosses the x-axis. The roots are the solutions to the equation.

The solutions are −2 and 2.

You can also tell which way a parabola will open by looking at the term containing the squared variable.

- If the term is positive, the parabola opens upward.
- If the term is negative, the parabola opens downward.





Key Terms

- completing the square
- parabola
- vertex
- zero property of multiplication

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Lesson 3: Quadratic Equations

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